Russell Tabata

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Education

- **University of Toronto** — Toronto, Canada
  *Bachelor of Applied Science - **Computer Engineering + PEY Co-op*** — *Sept 2025 - April 2030*
  ***Courses:** Engineering Strategies And Practices, Linear Algebra, Computer Fundamentals, Electrical Fundamentals*
- **Relevant External Coursework**
  Courses: Computer Architecture (Nand2Tetris), C Programming (Harvard CS50), Python Specilization (University of Michigan)

Skills Summary

- **Languages**: **C, HDL**, Assembly, Python, C++, Bash, Dart
- **Platforms**: Linux, Windows, Raspberry Pi, Git, AWS
- **Tools**: I2C, SPI, UART, CAN, STM32, Soldering

Experience

- **U of T Formula Racing Team — Firmware Team Member** — In-Person
  University of Toronto — *Sept 2025 - Present*
  - Selected for the firmware division; currently undergoing intensive training on STM32 microcontroller architecture and embedded C programming.
  - Learning industry-standard protocols (CAN bus) and Real-Time Operating System (RTOS) concepts to contribute to the 2026 vehicle control unit.
- **Tabata Systems — Lead Software Developer** — Remote
  Self-Employed (Contractual) — *Jun 2023 - Jan 2025*
  - Engineered and deployed scalable cloud infrastructure using AWS Lambda and Python to automate data processing workflows for enterprise clients.
  - Managed full project lifecycles for freelance clients, gathering requirements and delivering robust API integrated applications to calculate product quotes 100% more efficiently.
  - Collaborated with executive team on subscription model pricing and requirements

Projects

- **16-Bit Computer Architecture & CPU Design**
  HDL, Assembly, Digital Logic — *Sept 2025 - present*
  - Designed a fully functional 16-bit computer architecture (Hack platform) from the ground up, implementing elementary logic gates, adders, and flip-flops using Hardware Description Language (HDL).
  - Architected core hardware modules including the Arithmetic Logic Unit (ALU), CPU, and System Memory (RAM), validating data paths and control logic through rigorous hardware simulation suites.
  - Bridged the hardware-software interface by writing a custom assembler to translate symbolic assembly language into machine binary executable on the custom processor.
  - Gained practical understanding of practices and design tradeoffs relevant to modern CPU architectures.
- **iPod Retrofit & Embedded Controller** — Links: Website, Adafruit, Ext. Article
  C, GPIO, Raspberry Pi — *July 2024 - Aug 2024*
  - Integrated modern Raspberry pi hardware into a 2004 chassis, modifying internal wiring and power delivery to meet strict spatial constraints.
  - Engineered a custom C driver to interface the original Apple click-wheel with a Raspberry Pi via GPIO, utilizing bit-masking and debounce logic for precise input control.
  - Developed a state-machine based input system to map analog wheel signals to digital game controls, optimizing for $< 10ms$ input latency.
- **ViSTA - Vibro-Sensory Travel Aid Headset for the Blind** — Links: Website, Ext. Article
  Raspberry Pi, Python, Mobile Development, Computer Vision — *Sept 2024 - Oct 2024*
  - Designed a low-cost ($< \$100$) navigational aid using Python and OpenCV, achieving 54ms latency via an optimized ad-hoc communication protocol.
  - Offloaded intensive computer vision processing to mobile devices to minimize power consumption and thermal throttling on the headset unit.
  - Collaborated with CNIB for user testing, iterating on physical form factor based on feedback regarding weight distribution and vibrational intensity.

Honors and Awards

- Rick Hansen Difference Maker of the Year - May, 2025
- Charles Lee Design for Innovation Award - March, 2024
- Toronto Science Fair Gold Medalist - 2022, 2023, 2024